CHAPTER 7 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

Deep Learning and Agentic AI for UAV Cyber–Physical Security: Attacks, Detection, and Autonomous Countermeasures

AI-Driven UAS Cyber–Physical Security

The same flight symptom can come from a deliberate attack, a hardware fault, or the operating environment.

GNSS manipulation	A false position can pull the aircraft off route. Compare satellite navigation with an independent motion estimate before isolating a source.
C2 interference or replay	Loss bursts can trigger a link failsafe. Stale or unauthorized commands can change mission state. Check link health and message authenticity and freshness.
Sensor or perception corruption	A biased sensor or misleading camera input can hide an obstacle or distort pose. Check agreement with other sensors and track model uncertainty.
Compromised mission data	A modified waypoint or map can route the UAV outside its approved corridor. Verify provenance and validate geometry against mission limits.

Keep separate records for the observed anomaly, the suspected cause, and the action authorized for the aircraft.

AI for Navigation and Sensor Attack Detection

A temporal model can detect patterns across sensors. A physics-based baseline shows whether it adds value.

Align the evidence

Use timestamped GNSS, IMU, visual–inertial odometry (VIO), and link-quality data. Include sample age and missing-data flags in each window.

Establish a baseline

Compare each measurement with its predicted value. Use normalized innovations to account for uncertainty, then require persistence before raising an alert.

Compare a deep model

Train a temporal CNN on labeled windows or an autoencoder on nominal windows. Tune on validation flights and hold out entire flights for testing.

Compare event recall at the same false-alarm rate. Report detection delay separately for each anomaly type.

Agentic AI for Autonomous Countermeasure Selection

The agent proposes a response from an approved set. An independent supervisor controls execution.

Evidence
Alert, source health, age and uncertainty

→

Agent
Rank actions and attach evidence IDs

→

Safety supervisor
Check prerequisites and command expiry

→

Flight controller
Execute the authorized mode and confirm it

Mode prerequisites	Check localization integrity, geofence clearance, energy reserve, and the safe operating limits of the candidate mode.
Timeout or rejection	Invoke the configured fallback for the current state. Confirm the resulting mode and require sustained healthy data before recovery.

Return-to-home requires a trusted global position and home point. A hover also requires a usable local estimate.

Adversarial Machine Learning Against UAV AI Systems

The detector and the agent introduce attack surfaces of their own, from training data to mission-time inputs.

Evasion during flight	A manipulated input can suppress a detection or create a false one. Test plausible input changes and cross-check against independent sensing.
Training-data poisoning	Corrupted logs or labels can bias the detector. Track dataset provenance, review label changes, and test on an independently curated set.
Model or update tampering	A compromised artifact can change behavior after deployment. Verify signatures and approved versions, then test rollback to a known model.
Indirect prompt injection	A text-enabled agent may read instructions embedded in a log or payload. Treat external text as data and restrict the agent's tool permissions.

Evaluate the effect on the mission as well as the model score. A false alert can trigger an unnecessary mode change.

Case study — GNSS drift during dual-anchor UWB navigation

In simulation, one target UAV navigates with an IMU and ranges to two moving ultra-wideband (UWB) anchors.

Detect disagreement	Compare GNSS with predicted motion and UWB ranges. Label persistent disagreement as a navigation-integrity alert.
Check the UWB fallback	Check range and anchor-pose age, line-of-sight indicators, and geometry. Two ranges with known height can leave two horizontal solutions.
Resolve or abstain	Use a trusted motion prior to score the two branches. Keep both hypotheses if the evidence is weak, and reject degraded navigation if uncertainty exceeds its bound.
Choose an allowed mode	Slow down under healthy local aiding. If local aiding fails, use the prevalidated contingency for the available state and terrain.

Compare injected GNSS bias with benign UWB obstruction and stale anchor reports before assigning a cause.

Practical exercise — A controlled navigation integrity test

Fly the same simulated route under benign conditions, sensor corruption, and loss of the fallback source.

01	**Set up the test**	Use a 20 m route with 4 m waypoints and two moving UWB anchors. Log the truth trajectory and timestamps. Begin with nominal flights.
02	**Inject three cases**	At the 8 m waypoint, run separate cases with GNSS bias, UWB obstruction, and GNSS bias plus stale anchor poses. Apply changes to simulated inputs only.
03	**Compare responses**	Use the same scenario seeds for an innovation gate and a temporal detector. Run the same action policy, then compare agent proposals with approved modes.
04	**Produce evidence**	Submit a time-aligned plot of truth and estimated position, anomaly scores and alert times, plus a decision log of proposed and executed modes.

Repeat with held-out seeds and anchor trajectories. Fix thresholds and fallback prerequisites before the test runs.

Evaluation — Detection performance and mission outcomes

Score alerts by event and verify whether the response protected the aircraft within the available time.

Detection quality

Report detected attack events divided by injected events, false alerts per benign flight-hour, and median and 95th-percentile time from onset to alert.

Flight performance

Report horizontal RMSE, maximum position error, minimum geofence margin, and time from onset to confirmed contingency mode.

Supervisor behavior

Count rejected invalid proposals, missed deadlines, and unnecessary mode changes. Verify the response under stale or missing evidence.

Worked example

A bias starts at 12.0 s, the alert arrives at 13.2 s, and the approved mode is confirmed at 13.6 s. Detection takes 1.2 s and the full response takes 1.6 s.

Compare the full response time with the scenario’s recovery deadline. Report missed events separately from delay.

References and supporting material

GNSS spoofing	Kerns et al. (2014), Unmanned Aircraft Capture and Control Via GPS Spoofing. https://doi.org/10.1002/rob.21513
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Implementation	PX4 safety configuration and MAVLink 2 message signing. https://docs.px4.io/main/en/config/safety https://mavlink.io/en/guide/message_signing.html